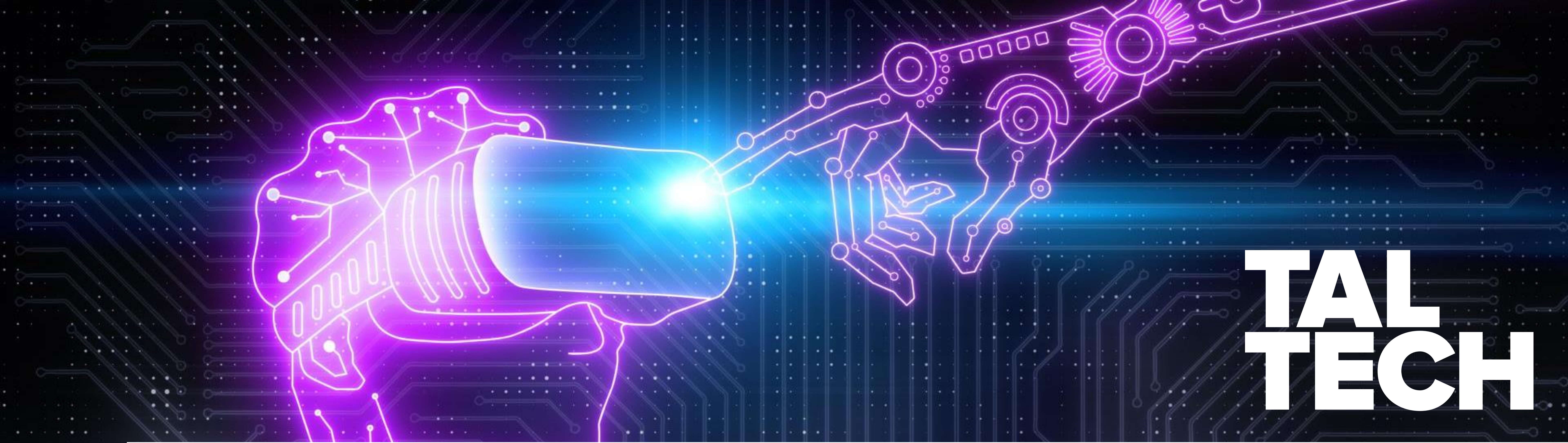

ITE4110: INFORMATION SOCIETY PRINCIPLES: TOWARDS E-GOVERNANCE

DR. ERIC JACKSON, JOSEPHINE LUSI

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E-GOVERNANCE TECHNOLOGIES AND SERVICES MASTER’S PROGRAM

A Bit About Me

- Dr. Eric Blake Jackson (just Eric)
- Grew up in Lincoln, Nebraska, United States
- International Relations background
- Started in 2014 at the e-Governance Academy in Tallinn
- Fulbright scholarship to the Republic of Georgia
- Have conducted interoperability workshops in Morrocco, Egypt, as well as Kenya
- Recently received my Doctorate in cross-border data exchange and interoperability
- Teach the Information Society Principles course

TAL TECH

COURSE BACKGROUND

- E-Gov 101: Building Blocks and Enablers
  - Zooming out
  - Estonian/European perspective
  - Global cases (GovStack)
- Pass/Fail Course (Over borderline 70 percent to pass)
- 6 ECTS
- Mixture of lectures, Moodle online learning materials/tasks, group discussion, and guest speakers + in-person November E-Governance Sprint Week
- Learner Composition:
  - Erasmus Students
  - E-Gov Master’s Students
  - Cybersecurity Master’s Students
  - EuroTeq Students

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COURSE SCHEDULE

- All assignments will be online and provided weekly in Moodle. Hybrid and in-person lectures will be conducted during the following dates:
  - 4th of September at 14:30 – 17:30; SOC-211b (Course Introduction, Information Society Principles, the Evolution of e-Governance Terminology)
  - 18th of September at 13:00 – 15:45; SOC-211b (Building Blocks of E-Governance, Estonian e-Governance Landscape)
  - 2nd of October 14:30 – 17:30; (Change Management, Strategy Orientation)
  - 16th of October 13:30 – 16:30; SOC-211B (Data Governance, Public Sector AI)

- **E-Governance Sprint Week**:
  - 5th of November 10:00 – 15:00; (Hands-on Session In-Person)
  - 6th of November 12:00 – 15:00; (Hands-on Session In-Person)
  - 7th of November 10:00 – 13:00; (Hands-on Session In-Person)
  - 8th of November 10:00 – 15:00; (Hands-on Session In-Person)

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COURSE LEARNING OBJECTIVES/OUTCOMES

- **After completing the course, the student:**

  - Can explain and analyze fundamental components of e-governance and their relation to information society;
  - Has an overview of the strategies and policies for governing digital transformation processes in the public and private sector;
  - Can assess and explain different technical, legal, political and social components of e-governance processes and components
  - Is capable of performing problem-based analysis on e-governance use-cases in written form;
  - Can present and convey a problem-based analysis to a larger audience

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GRADING/ASSESSMENT

- Continuous assessment is based on three components:
- 1) State of the Art Preliminary Case Study: 50% of the grade
- 2) Completion of all assignments of e-course in Moodle: 30% of the grade
- 3) Attendance of the lectures (in-person or online): 20% of the grade.
  - Mandatory In-person E-Governance Sprint Week (Nov. 5th – 8th)

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STATE OF THE ART PRELIMINARY CASE STUDY

- State-of-the-Art Preliminary Case Study: To pass the course, students are required to complete a State-of-the-Art Preliminary Case Study according to the following structure:
- 5 pages maximum (Single-Space), 12 Times New Roman, at least 10 academic + policy + grey literature references in IEEE format
- Template provided in Moodle and Grading Rubric in the Extended Syllabus
  - Introduction (.5 page)
  - Problem Statement (.5 page)
  - State of the Art (2 pages)
  - Discussion/Synthesis (2 pages)
  - Citations (.5 page)
- Due Date: December 8th

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STATE OF THE ART PRELIMINARY CASE STUDY OVERVIEW

- State of the Art Preliminary Case Study is the primary assessment component of the course.

- Case Selection and Research: Students select a case related to topics like data management, data governance, interoperability, AI applications, or digital ethics in the public sector or other digital transformation related topics.

- Tips for selecting a case:
  - Can be at a national level or European level
  - Can compare country x with country x
  - Can be a case related to the work you do as department/ministry/organizational leaders
  - Should be a problem-oriented case to explore and has literature sources around the problem topic.

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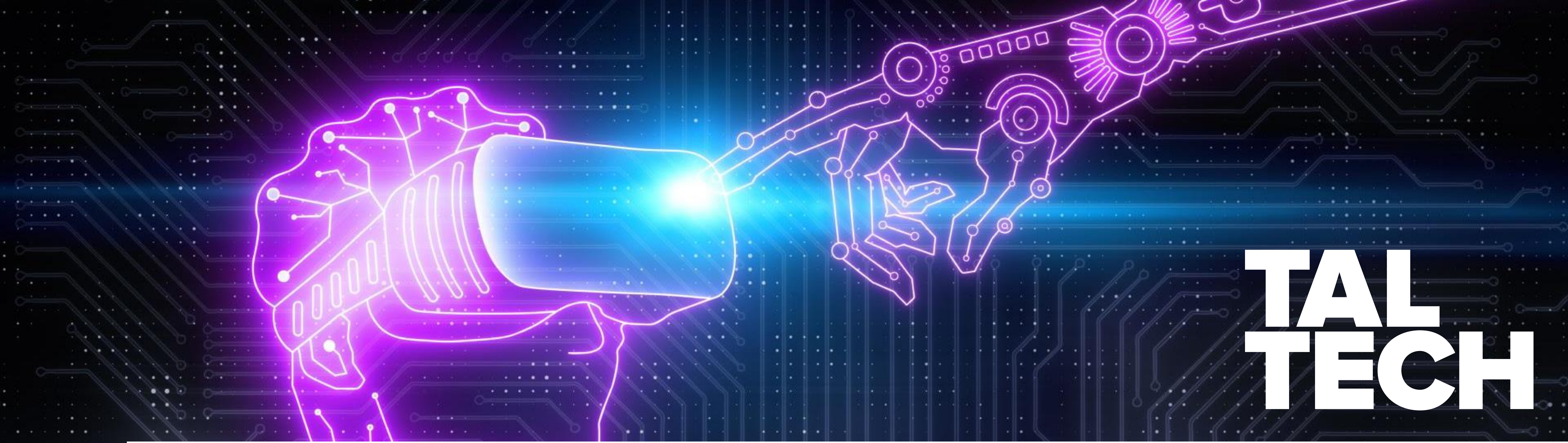
ONLINE MOODLE SELF-ASSIGNMENTS

Online: Topics and Assignments in Moodle

- Each week, students will watch online videos describing different facets of e-government and information society via Moodle
- Students will be required to watch the informational videos and completing the corresponding writing tasks.

Week of the course	**The current week's online assignment submission opens**	**Online assignment submission of the respective week closes: Deadlines**
Task 1: Sept. 4th – Sep. 11th	04.09	11.09 at 23:59
Task 2: Sept. 12th – Sep. 19th	12.09	19.09 at 23:59
Task 3: Sept. 20th – Sept. 27th	20.09	27.09 at 23:59
Task 4: Sept. 28th – Oct. 4th	28.09	04.10 at 23:59
Task 5: Oct. 5th – Oct. 11th	05.10	11.10 at 23:59
Task 6: Oct.12th – Oct. 19th	12.10	19.10 at 23:59
Task 7: Oct. 20th – Oct. 27th	20.10	27.10 at 23:59



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E-GOVERNANCE TECHNOLOGIES AND SERVICES MASTER’S PROGRAM

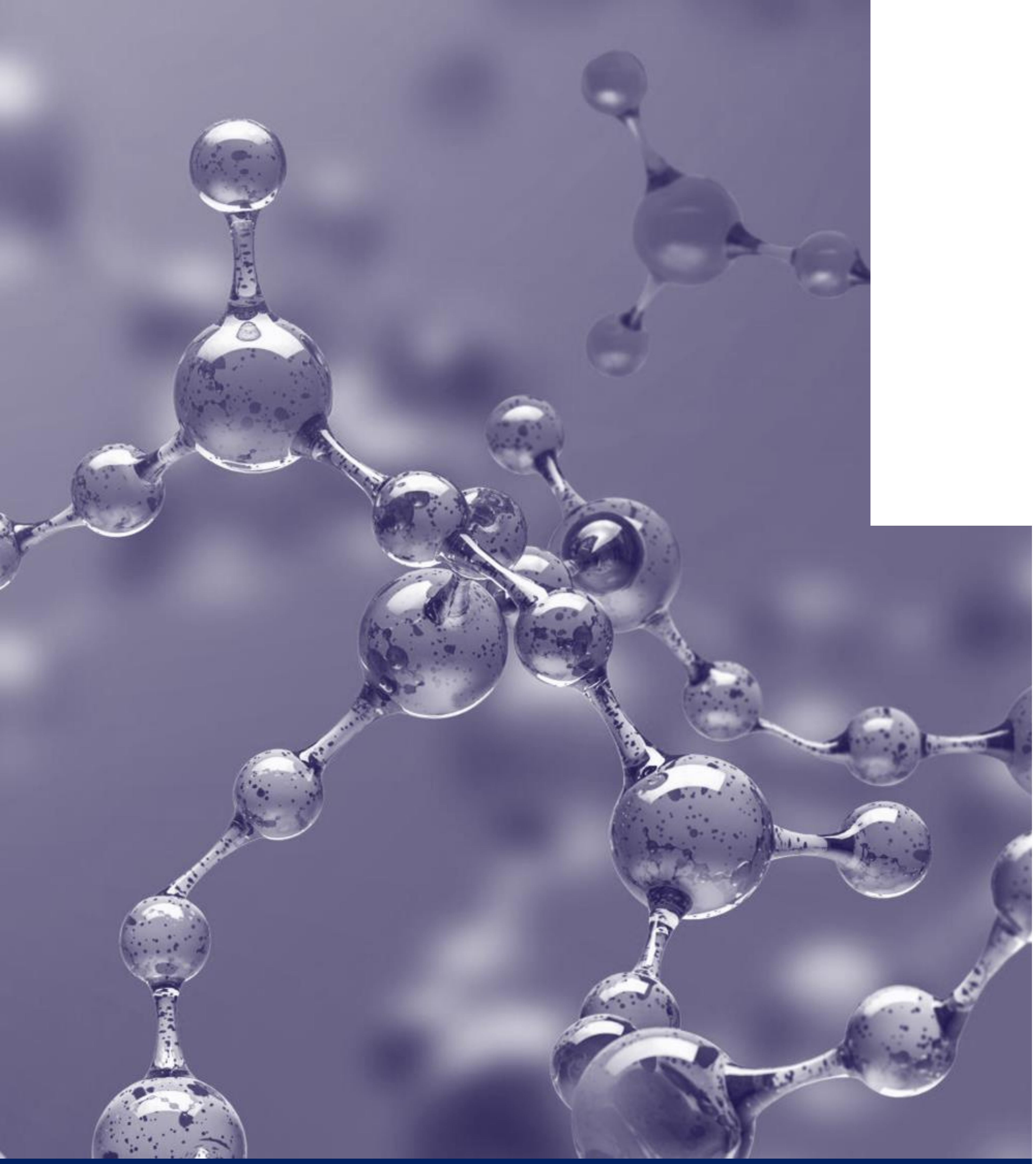
Scientists have measured the amount of data that enters the brain and found that an **average person living today processes as much as 74 GB** in information a day (that is as much as watching 16 movies), through TV, computers, cell phones, tablets, billboards, and many other gadgets”

(Frontiers for Young Mind’s, 2017)

“ Every year it is about **5% more than the previous year**”

“**Only 500 years ago, 74 GB of information** would be what a highly educated person consumed in a lifetime, through books and stories”

(Frontiers for Young Mind’s, 2017)



WHAT IS INFORMATION

What Is meaning of information in post-industrial context?

Semantic definition:	"Information is meaningful data (about something or someone) that may result from a systematic investigation" "Communication and reception are integral parts of information" "Information has effects"
In relation to information society, information's definition depends on the contextual lens we view it:	Technological: The transformation of information into computer readability and quantified into "bits" (binary, data structures) Economic: "data that have been organized and communicated" and is synonymous with the knowledge economy Cultural: Information overload has contributed to the loss of meaning and ability to make decisions - social media, TikTok, quick dopamine hits

WHAT IS INFORMATION SOCIETY?

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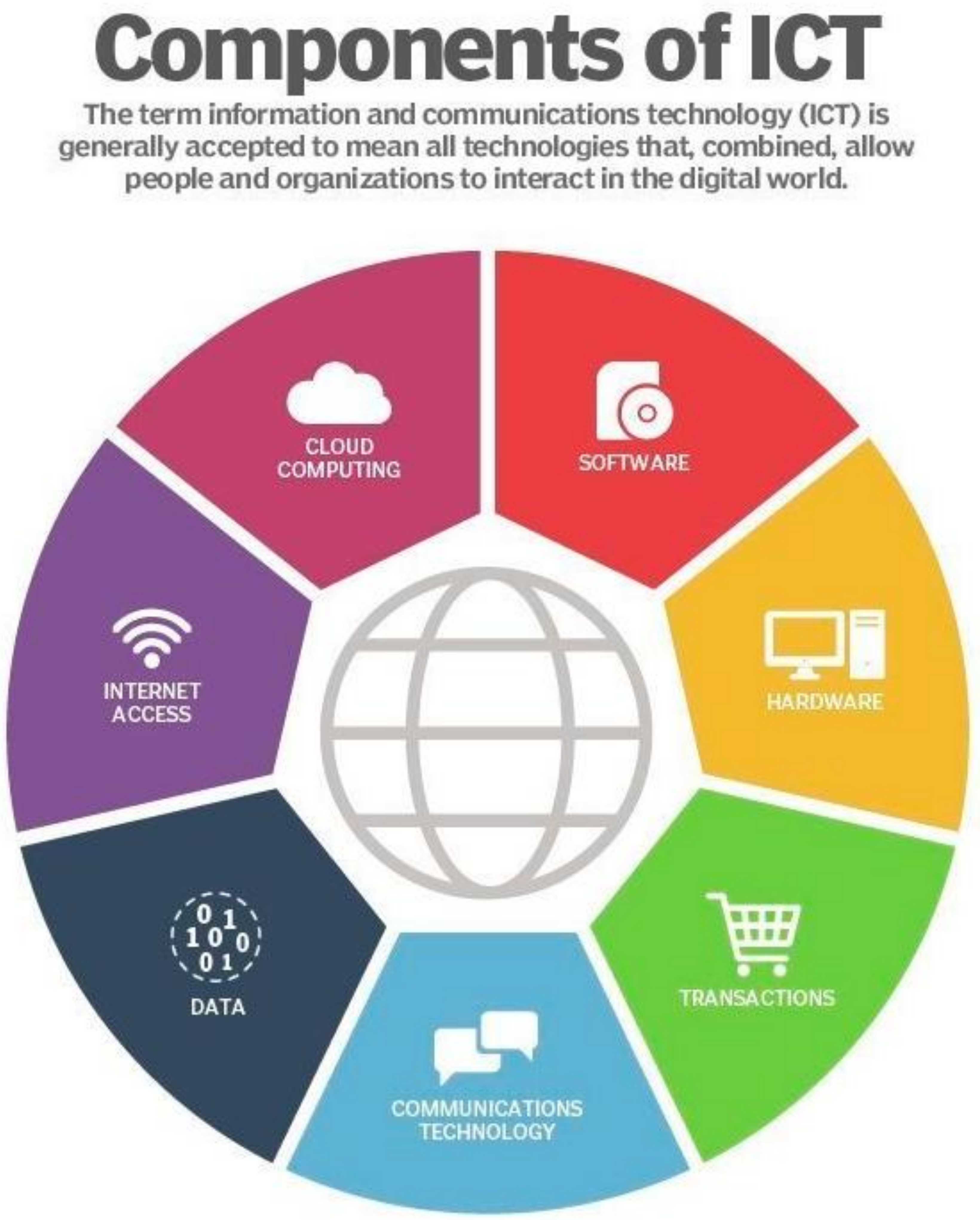
How to define it? It’s actually a trick question. (classic academic dispute)



Information Communication Technologies are central to Information Society

What Is Information Society?

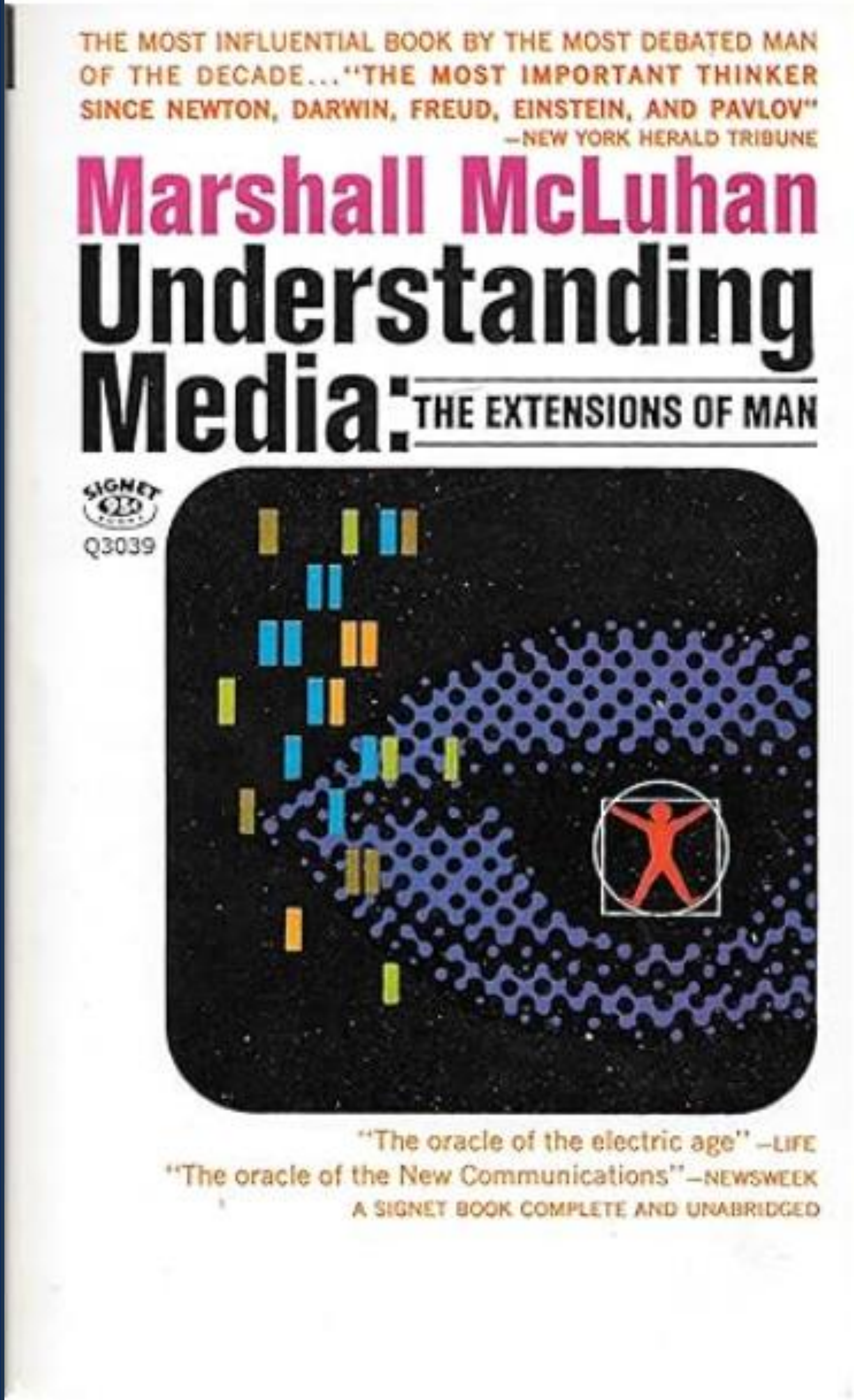
- Focus on ICTs, innovation in this area since the 1970s has drastically transformed society, economy, politics, war, governance etc.
- Has facilitated greater globalization due to ICTs' distributed, decentralized, and low-cost network capabilities
- Economic component: information economy, internet economy, knowledge economy
  - Separation of material and information.
  - It is not physical effort but “ideas, knowledge, skills, talent, and creativity” that create wealth in this new economy (Leadbeater, 1999).
- Sociological component: Created new social and power structures: white collar jobs, rural/urban divide, “information worker”
- Obliterated our conceptions of space and time:
  - Information superhighways: local, regional, national, and global information networks
  - Physical location does not matter anymore

Leadbeater, C. (1999). Living on thin air: The new economy. Viking/Penguin.

Characteristics of Information Society

Post-industrial context (knowledge economy, innovation)

- Intensive use and prioritization of Information Communication Technologies: computers, mobile phones, network infrastructure, IoT devices, etc. to mediate information
- Digitalization and increased ICT capability are rapidly transforming key areas where we organize and participate in society: education, economy, healthcare, government, and democracy, culture
- Network linkages: local, regional, national, and global information networks which imply interaction (communication and receiving)
- Processing, storing, computing, exchange of information growing exponentially.
- **Digital citizens - use computers and leverage decentralized network effects for facilitating communication, advocacy, and participation**



Marshall McLuhan: "The Medium is the Message"

- **Core Idea (1964, Understanding Media):** The form of a medium (TV, radio, print, internet) matters more than the content it carries.
- **Why?** Because media reshape how humans perceive, think, and interact, altering society and culture regardless of the "messages" transmitted.
- **Example:** The printing press (1440) standardized language and destroyed gatekeeping.
- **Example:** Television reshaped politics into visual performance (Kennedy vs. Nixon debates).

Applying McLuhan to the Internet

The Internet as Medium, not Just Channel

- It's not only about websites, emails, or posts (content).
- The structure of the internet (interactive, networked, decentralized) shapes how we experience information, authority, and community.
- The internet collapses distance → the experience of simultaneity becomes the message.

Hyperconnectivity and Social Behavior

- The medium of the internet fosters "network logic": communities form around affinity rather than geography.
- Content doesn't matter as much as the fact that people are always connected.
- The rise of memes, virality, and influencers shows that the way information spreads matters more than the actual message.

Information Abundance and Fragmentation

- The internet's affordance of endless choice shifts us from shared narratives (TV era) to fragmented micro-publics.
- The "message" is not just in what we read, but in the experience of personalization, algorithmic curation, and echo chambers.

From Consumer to Prosumer

- The medium blurs the line between producer and consumer (blogging, YouTube, TikTok).
- The message of the medium is that anyone can publish and participate.

The Internet as Extension of Human Faculties

- McLuhan described media as "extensions of humans" (wheel extends legs, book extends memory, phone extends voice).The internet extends memory (Google), identity (social media profiles), sociability (messaging apps), and cognition (AI tools).
- The message: our perception of self and society is inseparable from our digital extensions.

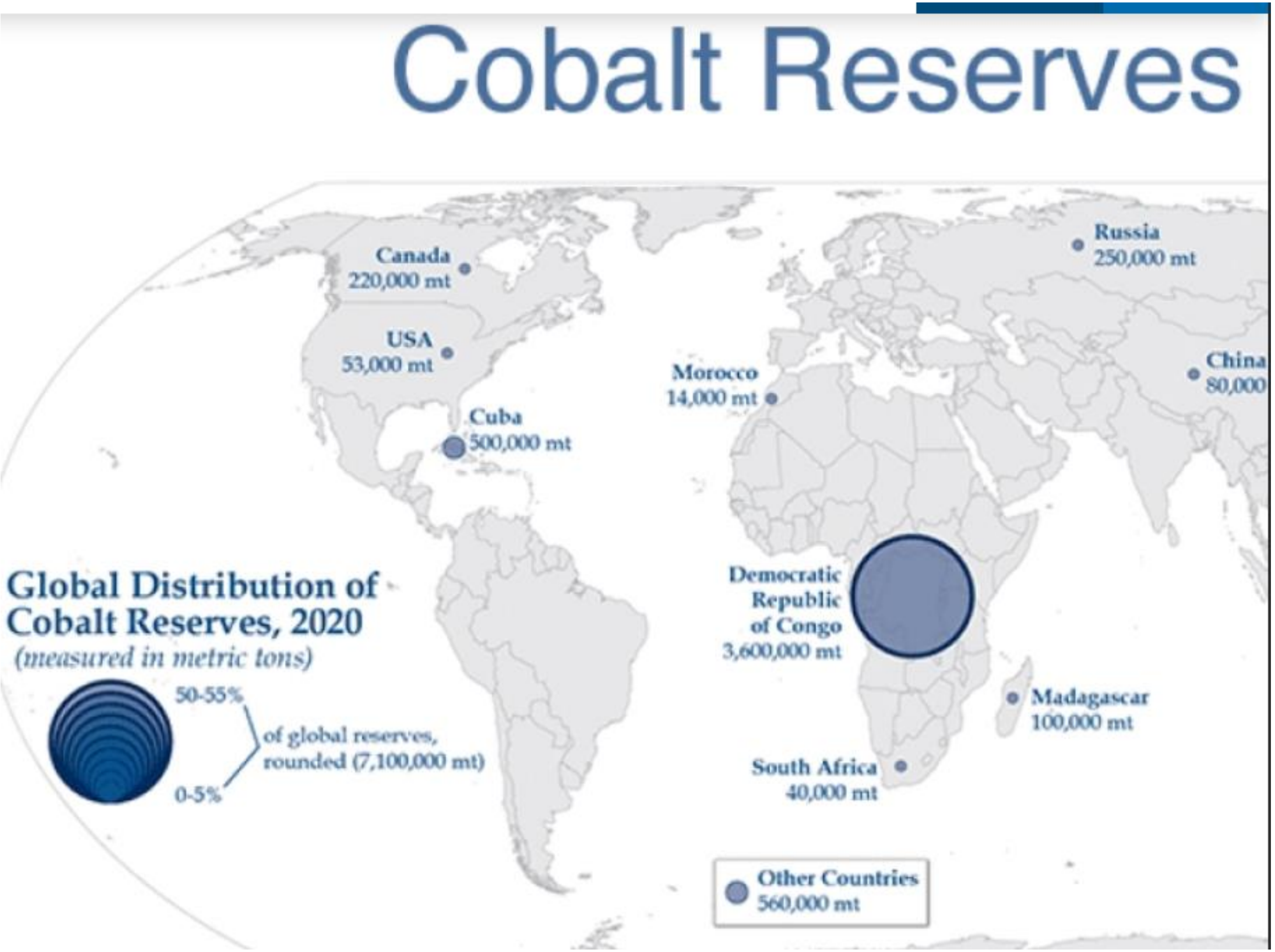
Principles of An Information Society

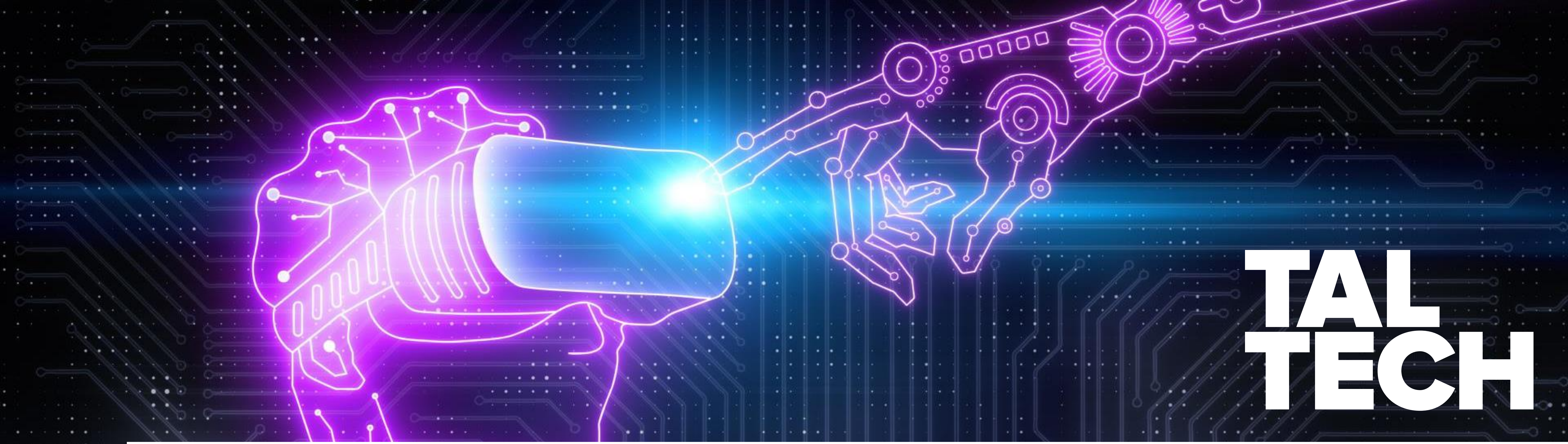
- Fair, equal, and accessible access to information = more robust information society.
- Digital literacy - how effectively are citizens able to use the internet? (Estonian Tiger Leap, AI-Leap)
- Bridging the digital divide - telecommunications infrastructure and connectivity
- Privacy and data protection - is it even possible nowadays?
- Open source/open data - transparent source codes, accessible and machine-readable data from the public sector
- Leverage ICTs to facilitate democratic participation - e-petitions, i-voting, social media have inspired revolutions (Arab Spring)

INFORMATION SOCIETY IN REALITY

- **Access to the internet limited for political reasons**
  - Great Chinese firewall
  - Freedom of online speech?
- **American techno-oligarchy** - Social media companies, centralized decision-making
- **Data protection** - is non-existent in some countries, Data is a key source of revenue in the information economy so it is highly sought out in various ways, some ethical, some not.
- **Disinformation and misinformation**
  - deep fakes
  - pandemic
  - war (Ukraine conflict)
- **Algorithmic black boxes (untrustworthy AI) and the rise of AI companies and Chip Manufacturers**
- **Crypto-economics, scammers, ponzis**
- **Supply chains for rare technological minerals often in exploited countries**

INFORMATION SOCIETY IN REALITY





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QUICK AGENDA

- Digital Society in brief and your thoughts
- E-Governance vs E-Government
- Digital Public Infrastructure vs Digital Public Goods
- Estonian Soft Power through DPI and the GovStack Initiative
- Questions

A Bit About Me

- Dr. Eric Blake Jackson (just Eric)
- Grew up in Lincoln, Nebraska, United States
- International Relations background
- Started in 2014 at the e-Governance Academy in Tallinn
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What Do You Picture When You Hear Digital Society?

DEFINING DIGITAL SOCIETY:

A society where digital tech and data are embedded in daily life and institutions, shaping communication, markets, governance, services, and participation.

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Characteristics of A Digital Society

- The proliferation of smart technologies and algorithms
- Datification of everything (IoT, Sensors, AI/ML)
- Networks and connectivity (5G, 6G)
- Platform mediation and the merging of consumer/producer
- Media literacy? Critical Thinking?

Principles of A Digital Society

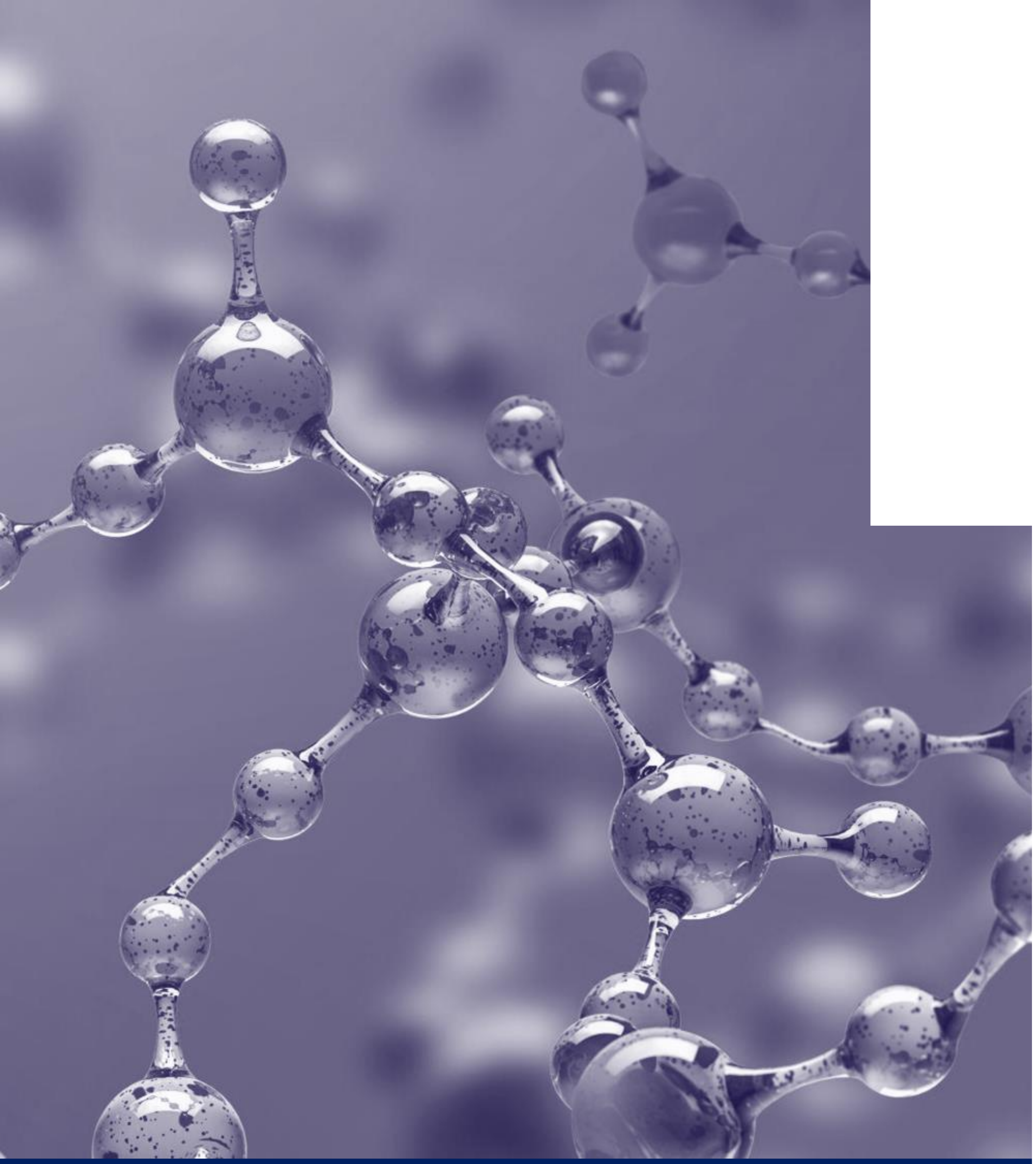
- Inclusion & accessibility
  - How to avoid digital exclusion/divide?
  - Affordability + skills
- Connectivity & reliability
  - Network infrastructure as a public-value enabler
- Trust, safety & cybersecurity
  - Safe participation
- Privacy & data protection
  - Rights + responsible data use
- Interoperability & standards
  - Different systems can “talk”; reduces friction
- User-centric services
  - Services designed around life events, not org charts
- Data governance & transparency
  - Accountability in data-driven decisions
- Resilience (continuity under outages, attacks, shocks)

Digital Society in reality: Patterns and Tensions

- **Reality patterns**
  - Mobile-first service delivery (citizen + consumer)
  - Platform mediation of daily life (attention, marketplaces, gig work)
  - Digital public services (IDs, registries, data exchange, life-event services)
  - Digital finance rails (payments + inclusion)
- **Tensions**
  - Digital divide becomes access + skills + usage + trust
  - Data power asymmetries (platforms/state/market)
  - Algorithmic opacity and legitimacy questions
  - Security and cascading failures

GROUP REFLECTION: What role does a society's culture and perception of trust play in an digital society?

GROUP QUESTION: What role does the public sector play in a digital society?



E-GOVERNMENT

VS

E-GOVERNANCE

What's the point?

- Frames how we look at digital government strategy and policy implementation
- Recognizing the differences between e-governance and e-government **enables governments to allocate resources to the areas that require the most attention**
- **Accurate measurement and evaluation.** Distinguishing allows for better metrics and assessment criteria tailored to the specific objectives of each concept.

E-government and E-Governance initiatives often involve multiple disciplines, including technology, public administration, policy, and social sciences. It becomes easier to foster interdisciplinary collaboration and expertise, ensuring that all relevant aspects are addressed effectively.

Before we begin...

- The definition of e-government and e-governance suffer from endless academic debates
- Various definitions: There is no unified definition of both concepts, but they have certain qualities that are distinguishable.

e-Government definitions

- “E-Government refers to the use by government agencies of information technologies (such as Wide Area Networks, the Internet, and mobile computing) that have the ability to transform relations with citizens, businesses, and other arms of government. These technologies can serve a variety of different ends: better delivery of government services to citizens, improved interactions with business and industry, citizen empowerment through access to information, or more efficient government management. The resulting benefits can be less corruption, increased transparency, greater convenience, revenue growth, and/or cost reductions.” - World Bank

- “the continuous optimization of service delivery, constituency participation, and governance by transforming internal and external relationships through technology, the Internet and new media.” - Gartner Group

- “E-government is defined as utilizing the Internet and the world-wide-web for delivering government information and services to citizens.” United Nations

Grigalashvili, V. (2022). E-government and E-governance: Various or Multifarious Concepts. *International Journal of Scientific and Management Research*, *5*(01), 183-196.

Characteristics of e-Government

E-government involves using information technology, and especially the Internet, to improve the delivery of government services to citizens, businesses, and other government agencies

E-government enables citizens to interact and receive services from the federal, state or local governments twenty-four hours a day, seven days a week.

Service-oriented and focused on the application of ICTs to deliver more efficient, secure, and quality services to citizens

Institution-based (different ministries/institutions provide different services with different objectives)

Different types of e-Government Relationships

G2C (Government to Citizens)

- Activities in which the government provides one-stop, online access to information and services to citizens.
- G2C applications enable citizens to ask questions of government agencies and receive answers; file income taxes (federal, state, and local); pay taxes (income, real estate); renew driver's licenses; pay traffic tickets; change their address; etc.
- e-Identity, e-service provision, e-Signature

G2B (Government to Business)
B2G (Business to Government)

- Focuses on the digital interactions between the government and businesses.
- It involves providing online platforms and services that facilitate business registration, licensing, permits, tax filing, procurement processes, and other government-business interactions.
- G2B e-governance aims to simplify administrative procedures, reduce red tape, and improve the ease of doing business.

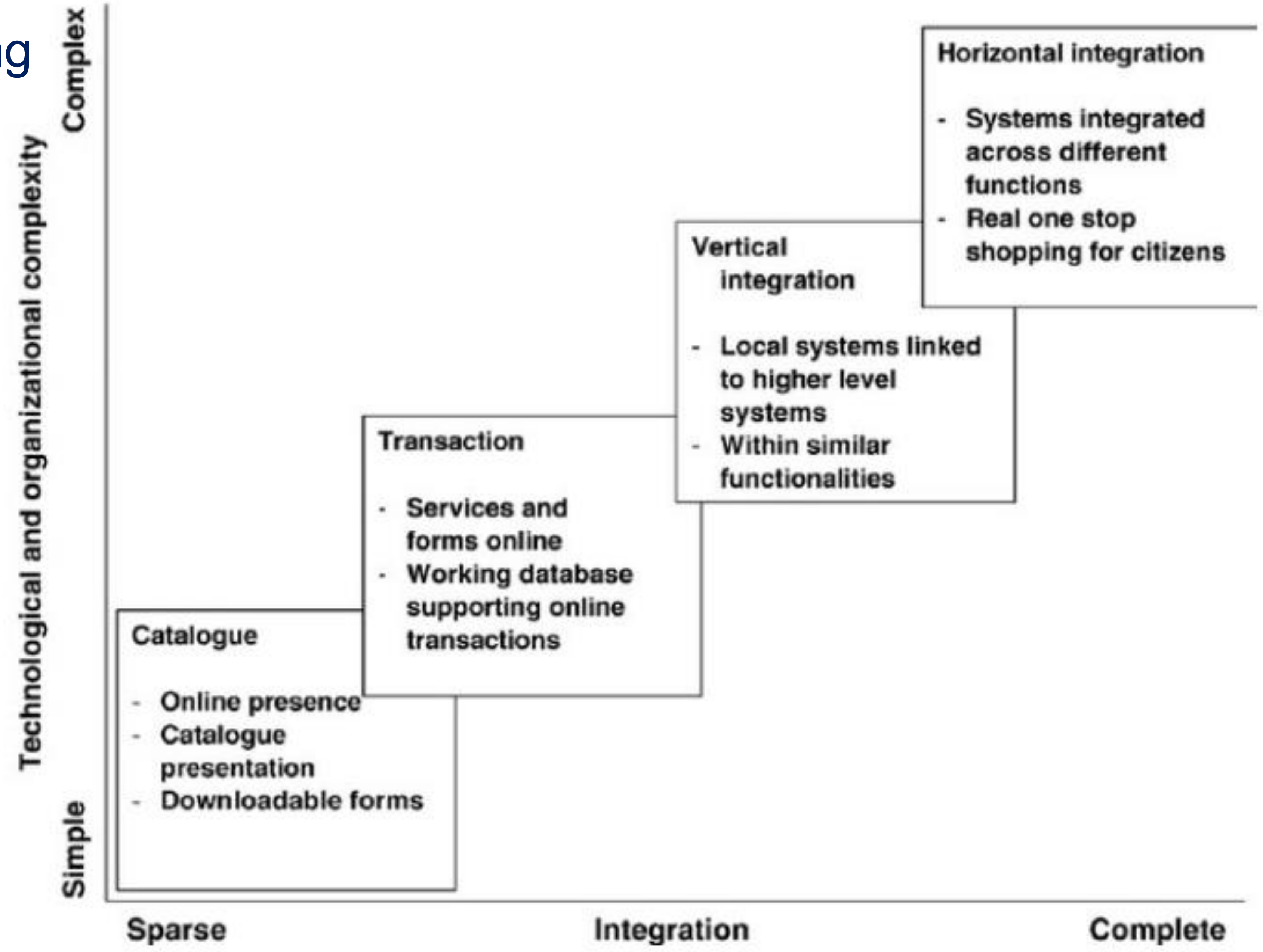
G2G (Government to Government)

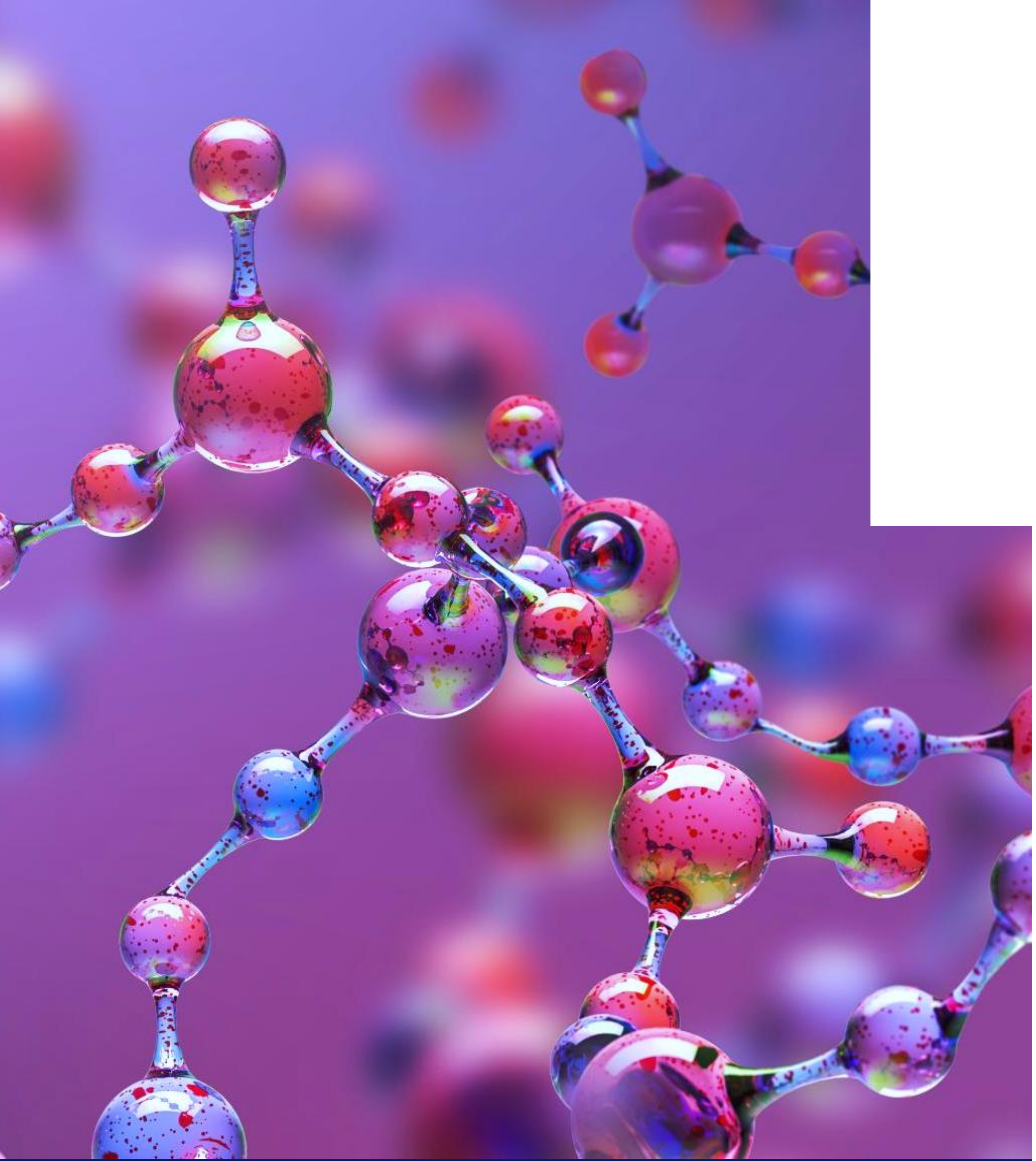
- G2G deals with those activities that take place between different government organizations/agencies.
- Many of these activities are aimed at improving the efficiency and effectiveness of overall government operations.
- Data exchange, database connectivity, data integration, data governance
- Internal IT infrastructure

e-Government Phases/Maturity Models

- Need a way of measuring, quantifying, and understanding e-government capabilities
  - Plethora of maturity models out there
  - Layne and Lee model

- Need a way of measuring, quantifying, and understanding e-government capabilities
  - Plethora of maturity models out there
  - Some other indicators:
    - The Telecommunication Infrastructure Index
    - Human Capital Index





E-GOVERNANCE

Governance definition

"Governance implies the processes and institutions, both formal and informal, that guide and restrain the collective activities of a group. Government is the subset that acts with authority and creates formal obligations. Governance need not necessarily be conducted exclusively by governments. Private firms, associations of firms, nongovernmental organizations (NGOs), and associations of NGOs all engage in it, often in association with governmental bodies, to create governance; sometimes without governmental authority."

Keohane, R. O., & Nye, J. S. (2000). In Nye, JS and Donahue, JD (editors), Governance in a Globalization World.

e-Governance definitions

- e-Governance is using information and communication technologies (ICTs) at various levels of the government and the public sector and beyond, for the purpose of enhancing governance" (Bedi, Singh and Srivastava, 2001; Holmes, 2001; Okot-Uma, 2000).

- "E-governance is the public sector's use of information and communication technologies with the aim of **improving information and service delivery, encouraging citizen participation in the decision-making process and making government more accountable, transparent, and effective**. E-governance involves new styles of leadership, new ways of debating and deciding policy and investment, new ways of accessing education, new ways of listening to citizens and new ways of organizing and delivering information and services.

- **E-governance is generally considered as a wider concept than e-government,** since it can bring about a change in the way citizens relate to governments and to each other. E-governance can bring forth new concepts of citizenship, both in terms of citizen needs and responsibilities. **Its objective is to engage, enable and empower the citizen"**

Grigalashvili, V. (2022). E-government and E-governance: Various or Multifarious Concepts. *International Journal of Scientific and Management Research*, *5*(01), 183-196.

Characteristics of e-Governance

- The overarching strategies, policies, decision-making arrangements, and processes an organization (public or private sector) adopts for leveraging ICTs to enable better governance on a local, regional, or national level through citizen participation and consultation
  - Involves improving service delivery and government functionality but from a broader perspective than just technology
  - Leadership
  - Policies, procedures, and standards
  - Legal environment
  - Financial allocation
  - Assesses the impact technology has on societal development
  - Long-term strategy (Estonia's Digital Agenda 2030)

Governance Models

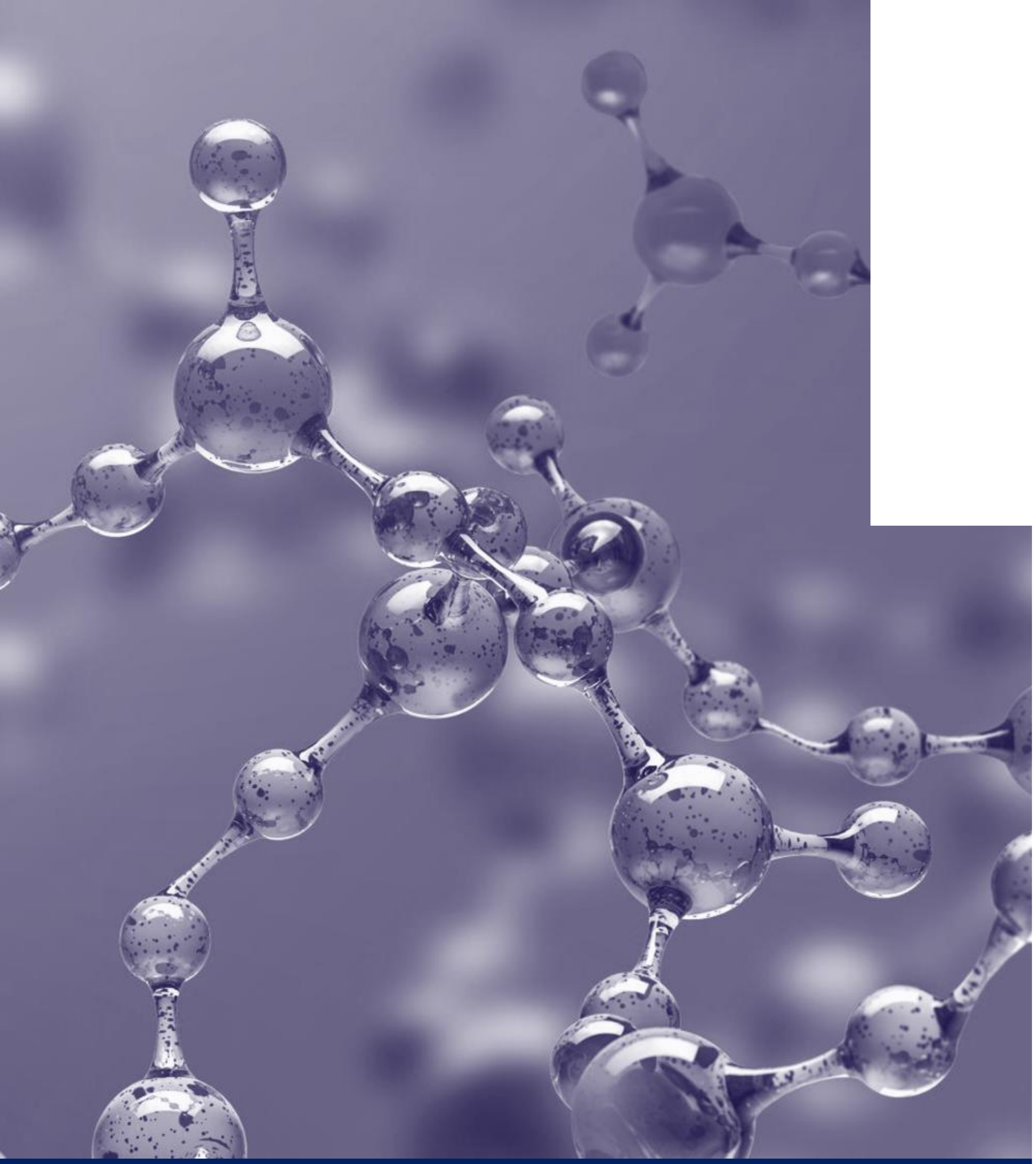
- CIO model
- Digital Ministry model
- Federated system
- No leadership

Comparison

	E-Governance	E-Government
Basic definition	A totality of technically mediated formal and informal interactions in which public and private actors participate in the policy design, development, and implementation.	Use of ICT to enhance efficiency, increase quality of services and empower citizens to participate in democratic process.
Purpose	Provides an opportunity to actively engage citizens in the political process.	Provides better services to citizens
Influence	How decisions are made	How decisions are carried out
Agenda	Relationship agenda	Service agenda
Projects	Shared databases, one-stop government	E-tax, e-transportation, e-health, driving license

- **e-Governance** is the overarching strategies, policies, standards, decision-making arrangements, and processes an organization (public or private sector) adopts for leveraging ICTs to enable better governance on a local, regional, or national level through citizen participation and consultation

- **e-Government** is more technically focused on the application of ICTs for making more efficient, transparent and accountable public services. This involves specific portals, application clients, etc. provided by specific institutions



OTHER E-GOVERNANCE TERMINOLOGIES

Whole of Government (WoG)

The Whole of Government (WoG) Approach is often referred to as joined-up government.

- „Whole-of-government denotes public services agencies working across portfolio boundaries to achieve a shared goal and an integrated government response to particular issues. Approaches can be formal or informal. They can focus on policy development, program management, and service delivery[1].“

- WoG is defined as an approach ‘in which public service agencies work across portfolio boundaries’ to develop integrated policies and programmes towards the achievement of shared or complementary, interdependent goals[2].

1. Australian Management Advisory Committee’s Connecting Government report (2004)
2. Christensen, T., & Lægreid, P. (2007). The whole-of-government approach to public sector reform. Public administration review, 67(6), 1059-1066.90

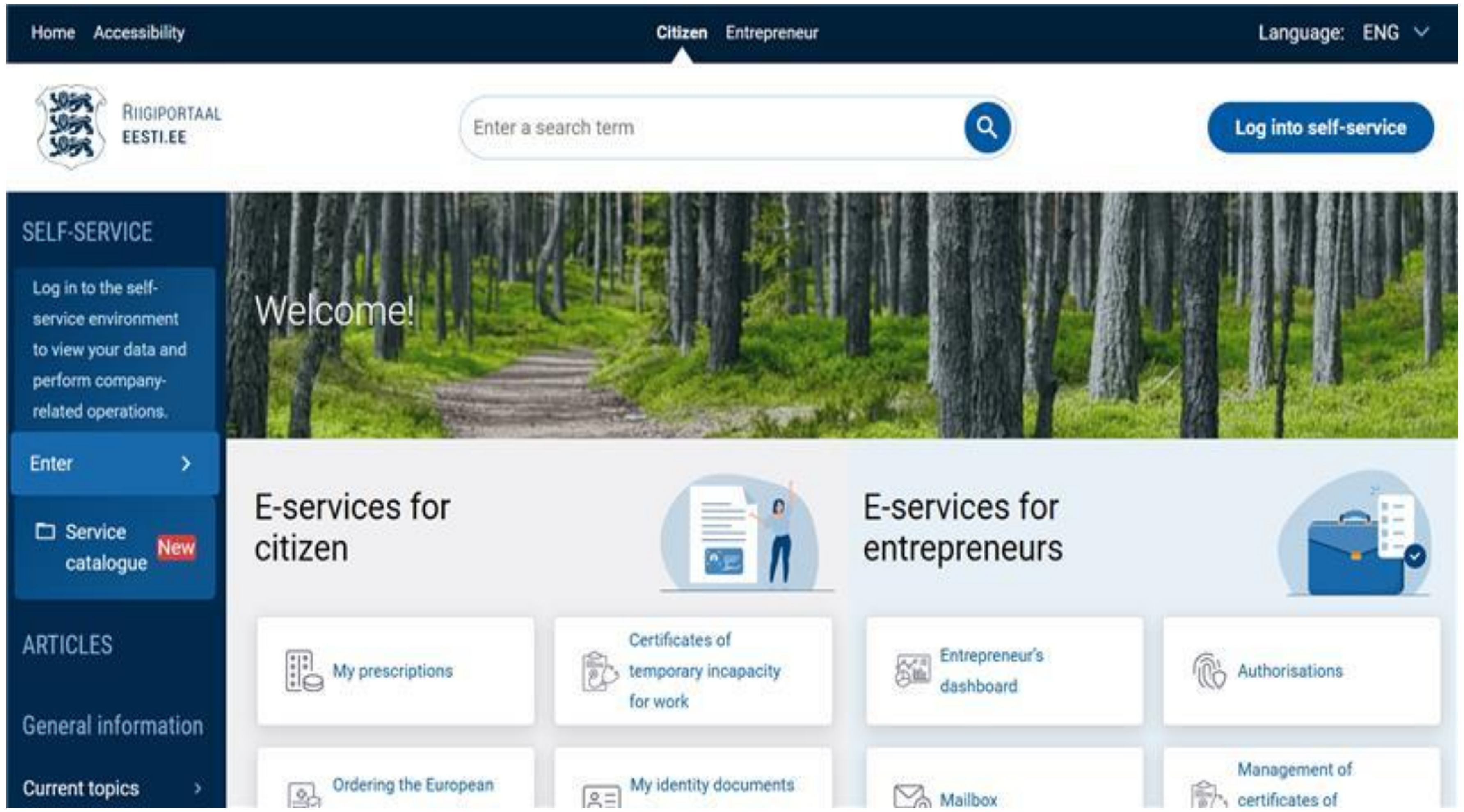
Government as a Platform (GaaP)

- „Government stripped down to the essentials. A platform provider builds essential infrastructure, creates core applications that demonstrate the power of the platform and inspire outside developers to push the platform even further, and enforces “rules of the road” that ensure that applications work well together[1].“

- „The concept of Government as a Platform (GaaP) envisages the transformation of the coordination among all national and local public agencies which compose public administration, from closed, structured, and formalised hierarchical relationships into open, flat, and unstructured relationships. Shared software, data, and services enable coordination among public agencies and open the public service production processes to actors who have been traditionally external to public administration[2].“

1. O’Reilly, T. (2010). Government as a Platform. In: D. Lathrop eds., Open Government: Collaboration, Transparency, and Participation in Practice, Sebastopol, Calif.: O’Reilly Media, pp.13-40.
2. Cordella, A., & Paletti, A. (2019). Government as a platform, orchestration, and public value creation: The Italian case. *Government information quarterly*, *36*(4), 101409.

Governance platforms in Estonia



GROUP REFLECTION: What is required to implement successful e-Governance

Estonian e-Governance at a Glance

- **Core Foundational/Enabling Technical Components:**
    - **e-Identification**
        - 99% citizen uptake as its compulsory
        - Authentication methods: Mobile-ID, Smart-ID, Physical ID
    - **Public Key Infrastructure Model**
        - Citizens have public key (National identity number) and two private keys
    - **Digital Signature**
        - Hand-written signatures equivalent to digital signatures by law
        - 800 million+ digital signatures have been performed
    - **Interoperable Data Exchange (X-Road)**
        - X-Road data exchange layer enables secure data exchange between G2G/B/C
        - Compulsory to use in public sector
        - **Consent mechanisms for data exchange and processing**
    - **Digital Payments – Not emphasized enough in Estonia**

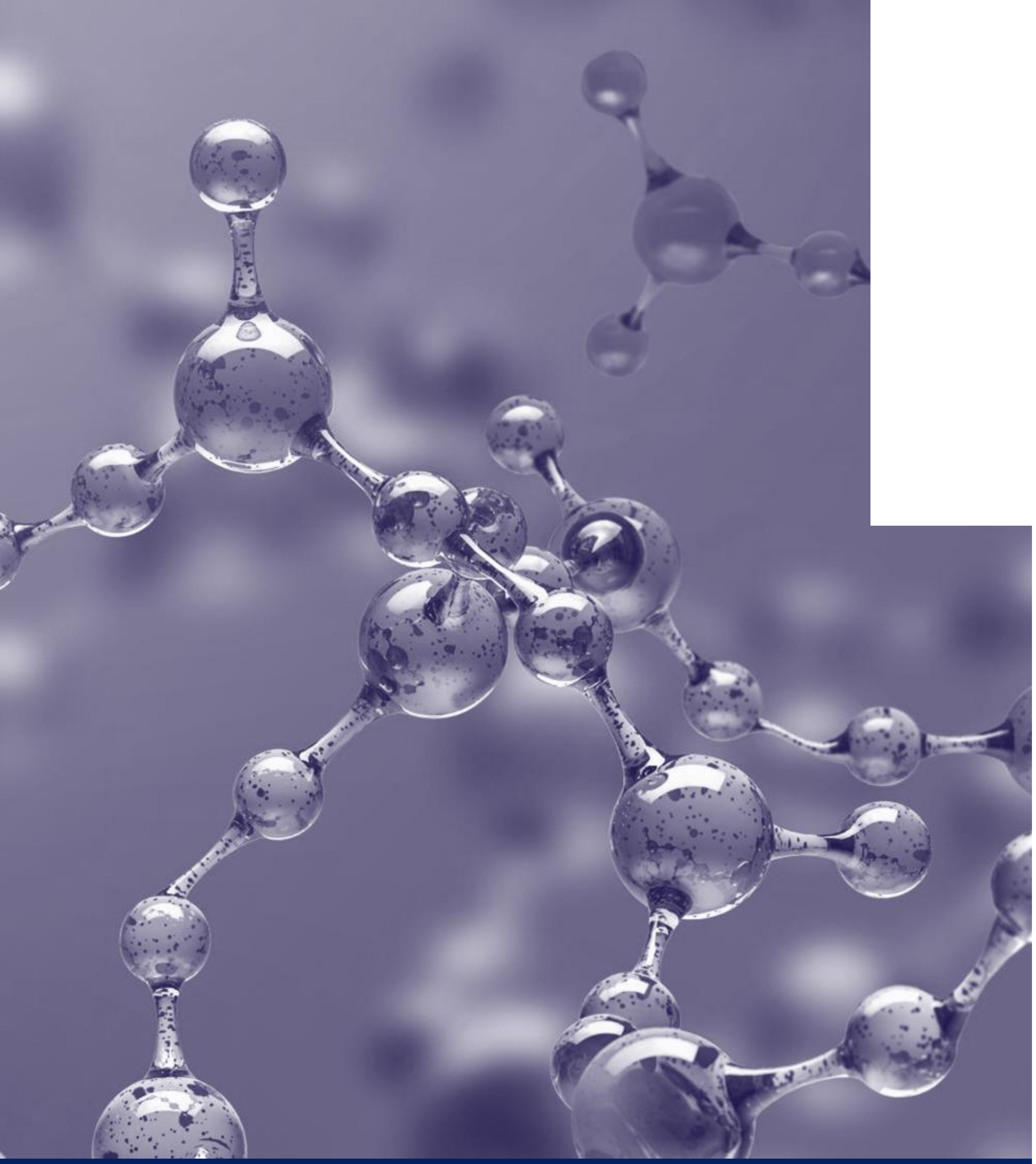
- **Examples of Estonian Enabler Outputs:**
    - **i-Voting**
        - Citizens have the right to vote through the internet (e-Democracy/Participation)
    - **e-Services**
        - eesti.ee is the citizen portal for accessing e-services, seeing which public authorities have accessed data, 3000+ e-services available 24/7, proactive
- **Synergy between legal frameworks, digital public infrastructure, and citizen perceptions**

Estonian e-Governance at a Glance

- Core Foundational/Enabling Estonian Governance Components:
  - Legal Frameworks
  - Estonian soft power in this field: influencing e-government solutions and initiatives
    - GovStack
    - Cybernetica
    - Nortal
    - E-Governance Academy

Estonian e-Governance at a Glance

- Core Foundational/Enabling Estonian Governance Components:
  - Private-Public Partnerships
    - Certificate authority is a private company (SK ID)
    - X-Road Data Exchange layer is owned by a non-profit (Nordic Institute for Interoperable Solutions) funded by the Estonian and Finnish Ministries of Finance
    - Who owns the data? How is this established?



DIGITAL PUBLIC INFRASTRUCTURE

Where does it fit in?

DPI Whole of Society (WoS) Approach

- A whole-of-society approach is of interest in complex problems that require more sustainable, proactive approaches that lie outside the mandate of any single department or agency, private organisations, and individual members of the public[1].

- WoS represents a broader approach, moving beyond public authorities and engaging ‘all relevant stakeholders, including individuals, families and communities, intergovernmental organizations, religious institutions, civil society, academia, the media, voluntary associations and […] the private sector and industry’[2].

1. Gonçalves, D. P. (2021). The concept of ‘system’ in a whole-of society approach.
2. Ortenzi, F., Marten, R., Valentine, N. B., Kwamie, A., & Rasanathan, K. (2022). Whole of government and whole of society approaches: call for further research to improve population health and health equity. *BMJ Global Health*, *7*(7), e009972.

Digital Public Infrastructure

We recognize **DPI**[1] as a set of shared digital systems which are **secure and interoperable**, built on **open standards and specifications** to deliver and provide **equitable access to public and/or private services** at societal scale and are governed by enabling rules to drive development, **inclusion,** innovation, trust, and competition and **respect human rights and fundamental freedoms**.

Digital (railroads)

1. Enables remote, paperless, presence-less service delivery
2. Reducing cost and increasing access through digital

Public

1. For the public interest
2. Public governance and accountability to people

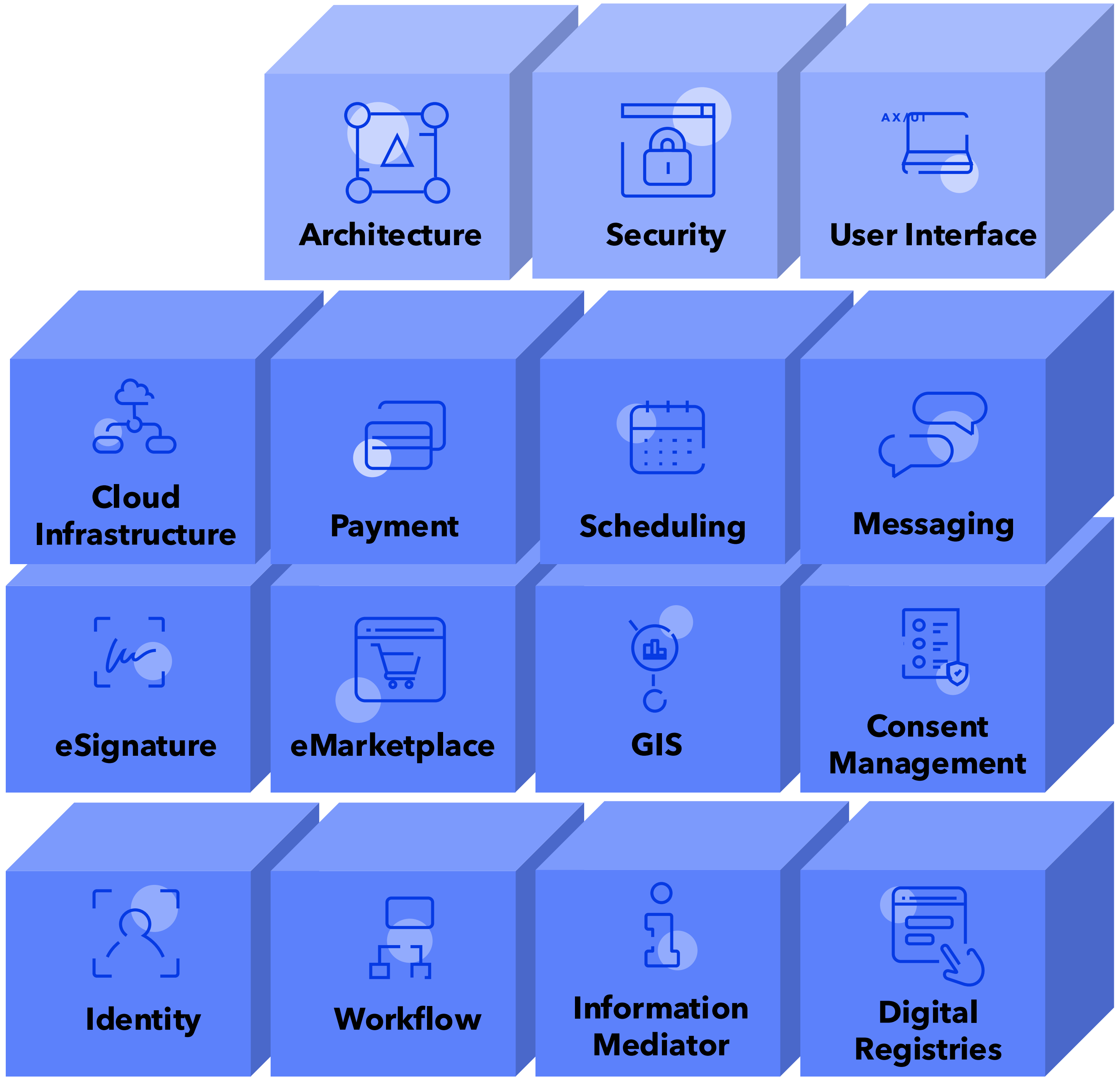
Infrastructure

1. Building blocks for large-scale development of digital solutions
2. Ecosystem-led implementation: private-sector led, public-private led, or public-sector led
3. Ecosystem-level impact: can be leveraged in public and private domains

1. Source: G20 consensus on DPI, under India's G20 Presidency, 2023 (refer to the Digital Economy Declaration for exact language). This language of DPI is based on reviewing different conceptions from World Bank, CDPI, DIAL, DPGA, OECD, Co-Develop, GovStack, and other organizations.

GovStack contributions

- ✓ **Technical Specification and Standards** for Building Blocks to build the basis for scalable, interoperable digital services
- ✓ **A safe digital testing environment (sandbox)** to learn, experiment, and prototype services and reference implementations
- ✓ Workshops, trainings and e-learnings for **capacity building** and skills development.



Republic of Estonia Ministry of Foreign Affairs

Bundesministerium für wirtschaftliche Zusammenarbeit und Entwicklung

digital impact alliance

DPI Conversations…

Other Leading Contributors

Technology Providers

mojaloop foundation

MOSIP

International Organisations

WORLD BANK GROUP

ITU

United Nations Office for Digital and Emerging Technologies

dial Digital Impact Alliance

UNDP

Philanthropic Foundations

CO-DEVELOP

BILL & MELINDA GATES foundation

Civil Society & Alliances

Centre for Digital Public Infrastructure

Digital Public Goods Alliance

Other Contributors

Private Institutions

Deloitte.

aws

Open-source community

Mifos Initiative

OSB Open Source Business ALLIANCE

iGrant.io Your data, your choice.

OpenWallet FOUNDATION

OSIA

Digital Public Infrastructure vs Digital Public Goods

Digital Public Infrastructure	• **Digital backbones and enablers of public service delivery** • Encompasses **digital ID, payments, and data exchange.** • **Includes both open source and proprietary solutions** • **Examples:** Aadhaar (proprietary), RAAST (proprietary), R-HMIS (open source) based on DHIS2, X-Road
Digital Public Goods	• **Open-source software, open data, open artificial intelligence models, open standards, and open content** that adhere to privacy and other applicable international and domestic laws, standards, and best practices, and do no harm • **Open source only** • **Examples:** MOSIP, DHIS2, MOJALOOP

Some DPIs are based on DPGs

Open source versus proprietary

- Open-source solutions can be freely adopted or adapted
- Proprietary solutions are owned by a government or private company and cannot be freely adopted or adapted

Source: Definitions proposed by the UNSG Roadmap for Digital Cooperation and the Digital Public Goods Alliance

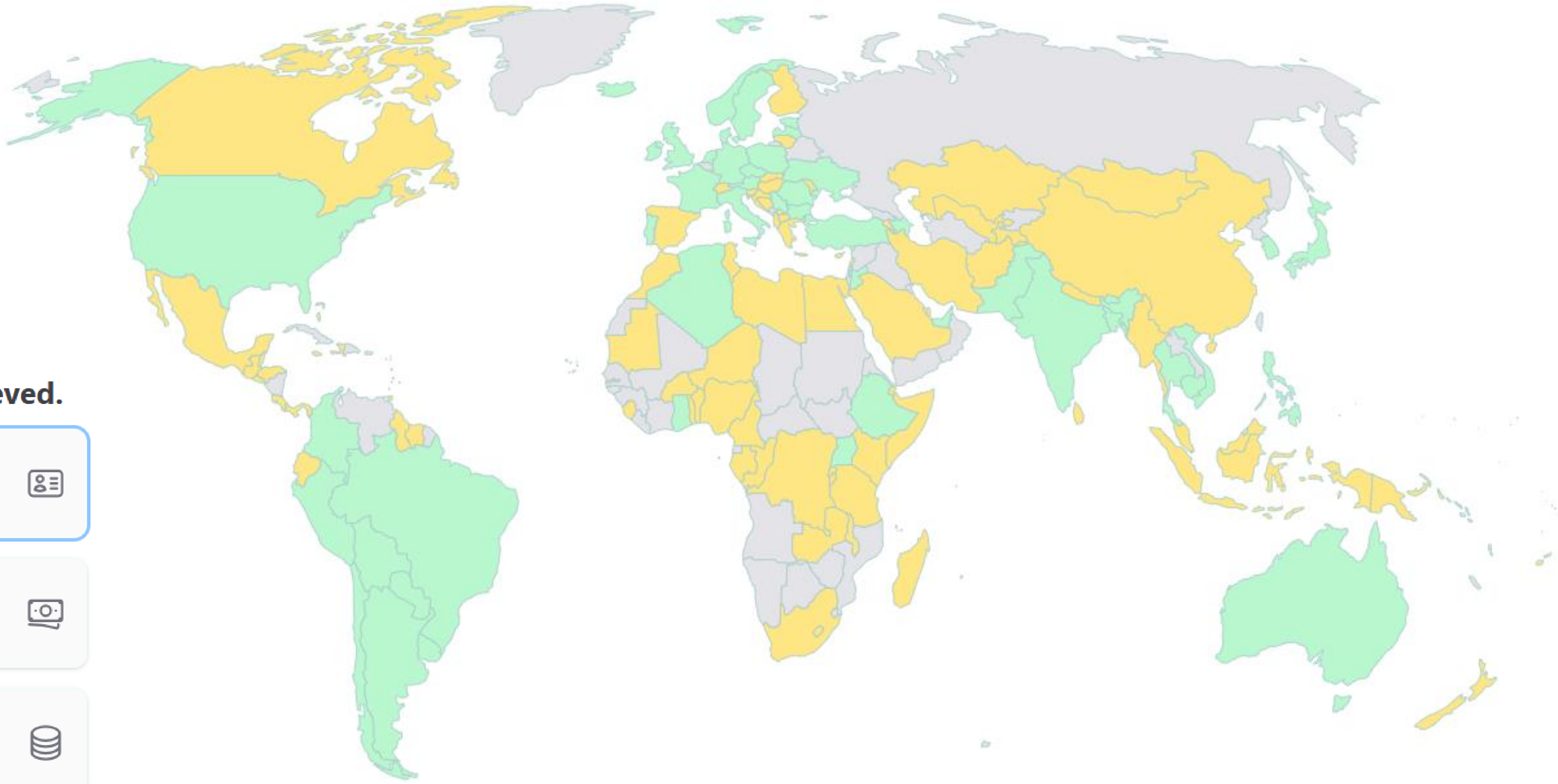
DPI MAP

DPI is far more prevalent than initially believed.

57 countries have a DPI-like
Digital ID System

100 countries have a DPI-like
Digital Payment System

102 countries have a DPI-like
Data Exchange System



Digital ID System implementation status by country

DPI Map (v2, 2025-03-31). Institute for Innovation and Public Purpose, UCL. https://dpimap.org/data

GROUP REFLECTION: Digital sovereignty vs open-source? What might be potential issues?

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